9749 H2 Physics 2025 GCE A-Level Suggested Solutions

Paper 1

1 **C** Considering $F = BIL$

Units of $F = (\text{Units of } B)(\text{Units of } I)(\text{Units of } L)$

$\text{N} = \text{TAm}$

$\text{T} = \dfrac{\text{N}}{\text{Am}}$

Since $\text{N} = \text{J m}^{-1}$ (from Work done = Force × displacement)

$\text{T} = \dfrac{\text{J}}{\text{Am}^2} = \text{J A}^{-1}\ \text{m}^{-2}$

2 **D** $a = \dfrac{bc^2}{d} \Rightarrow \dfrac{\Delta a}{a} = \dfrac{\Delta b}{b} + 2\dfrac{\Delta c}{c} + \dfrac{\Delta d}{d}$

$= \dfrac{3}{24} + 2 \times \dfrac{2}{13} + \dfrac{1}{18} = 0.4882$

$\therefore \dfrac{\Delta a}{a} \times 100\% \simeq 49\%$

3 **B** In the vertical direction, $s_y = \left(\dfrac{u_y + v_y}{2}\right) t$

$\therefore$ max height reached $= x = \left(\dfrac{u\sin\theta + 0}{2}\right) t$ ..... (1)

But horizontal displacement in time t is $(u\cos\theta)\, t = 3x$ ..... (2)

$\therefore \dfrac{\text{eqn}(1)}{\text{eqn}(2)} = \dfrac{\cancel{x}}{3\cancel{x}} = \left(\dfrac{\cancel{u}\sin\theta}{2\cancel{u}\cos\theta}\right)\dfrac{\cancel{t}}{\cancel{t}} \quad \Rightarrow \tan\theta = \dfrac{2}{3}$

4 **B** resultant force $=$ $ma = upthrust - weight$

$a = \dfrac{\cancel{V}\rho_w g - \cancel{V}\rho_{ice} g}{\cancel{V}\rho_{ice}}$, where V = volume of ice-cube

$= \dfrac{(1.0 - 0.92) \times 9.81}{0.92} = 0.853\ \text{m s}^{-2} \simeq 0.85\ \text{m s}^{-2}$

5 **C** At t = 30 s and 60 s, the velocity is constant, and there is no acceleration. Hence the resultant force at t = 30 s and 60 s are both zero.

6 **B** Initially, torque $T = F(3x) - Fx = 2Fx$

where x is the distance PQ (also = QR, RS and SM) and F = magnitude of F_1 & F_2

When $F \rightarrow \dfrac{F}{2}$ and F_2 has shifted to R,

new torque $= \dfrac{F}{2}(2x) - \dfrac{F}{2}x = \dfrac{F}{2}x = \dfrac{T}{4}$

7 **B** Initially, $W = kd$, where k is the force constant of the wire.

This wire is actually made up of 2 halves of the wire, each having an extension of $\frac{d}{2}$.

Hence, the equivalent force constant of each half is $2k$.

When the object is hung from the midpoint, the tension in each wire is $\frac{W}{2}$.

$$\frac{W}{2} = (2k)d_{new} \Rightarrow W = k(4d_{new}) \Rightarrow d_{new} = \frac{d}{4}$$

8 **A** For an elastic collision, relative velocity of approach = relative velocity of separation

$\therefore u - 0 = v_Q - v_P = v$

Since u = v, the graph is a linear graph with a constant positive gradient passing through the origin.

9 **B**

$$\text{Work done} = W \times s = P \times t \Rightarrow s = \frac{Pt}{W}$$

10 **A**

$$\text{Average speed} = \frac{\text{Total distance}}{\text{total time taken}} = \frac{2\pi(0.30)}{66 \times 60} = 4.8 \times 10^{-4} \text{ m s}^{-1}$$

11 **A** Since $g \propto \frac{1}{r^2}$, where r is the distance of the point from centre of Earth,

g decreases as r increases.

12 **C** Since P and Q are in thermal equilibrium, there is no net transfer of energy between them.

13 **A** $pV = nRT = nR(\theta + 273.15) = nR\theta + 273.15nR$

$\therefore$ gradient of graph of pV against θ is nR

$$\text{gradient} = \frac{150 - 135}{30 - 0} = \frac{1}{2} \quad \rightarrow \therefore n = \frac{1}{2 \times 8.31} = 0.060 \text{ mol}$$

14 **C** Internal energy of a body depends on both quantity of substance and temperature of the body, but average kinetic energy of molecules depends only on the temperature of the body.

15 **A** $\Delta U = Q + W = 50 - 30 = 20 \text{ J}$

16 **B** Since total energy in an undamped simple harmonic motion is a constant,

Total energy $= E_P + E_K$

$E_P =$ Total energy $- E_K$

17 **A** Option A is not an example of resonance because the oscillator (in this case, the loudspeaker) is not coupled to a driver that is oscillating at the same natural frequency as the loudspeaker. The volume is increased by supplying more (electrical) energy to the circuit containing the loudspeaker.

18 **B** The heaps of powder indicate the positions of the nodes of the stationary wave (where the air particles are not moving). Hence, the distance between adjacent heaps of powder is half a wavelength.

$$3.5\left(\frac{\lambda}{2}\right) = 3.5 \text{ m} \Rightarrow \lambda = 2.0 \text{ m}$$

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19 **D** The shortest time taken for the displacement at X to return to the original displacement of $-\frac{A}{2}$ is one period, T. Thus, the time taken for the displacement to become $\frac{A}{2}$ is $\frac{T}{2}$.

20 **D**

Since $F \propto \frac{1}{r^2}$,

$$\frac{F_f}{F_i} = \left(\frac{r_i}{r_f}\right)^2$$

$$\therefore F_f = \left(\frac{9.00\times10^{-3}}{1.00\times10^{-3}}\right)^2 \times 2.84\times10^{-24} = 2.30\times10^{-22}\ \text{N}$$

21 **D**

$$V_X = \frac{Q}{4\pi\varepsilon_0 r} = \frac{Q}{4\pi\varepsilon_0 \times 1.2\times10^{-5}} = -1.2\times10^{-4}$$

$\therefore$ the electric field strenth at the mid-point between charge Q and X is

$$\frac{Q}{4\pi\varepsilon_0\left(\frac{r}{2}\right)^2} = \frac{4}{r}\times\frac{Q}{4\pi\varepsilon_0 r} = \frac{4\times1.2\times10^{-4}}{1.2\times10^{-5}} = 40\ \text{NC}^{-1}$$ and is directed to the left.

The field strength at the same mid-point due to the α-particle is

$$\frac{2e}{4\pi\varepsilon_0\left(\frac{r}{2}\right)^2} = \frac{2\times1.6\times10^{-19}}{4\pi\times8.85\times10^{-12}\left(0.6\times10^{-5}\right)^2} = 79.9 \simeq 80\ \text{NC}^{-1}$$, directed leftwards.

$\therefore$ resultant electric field strength at mid-point is $40 + 80 = 120\ \text{NC}^{-1}$

22 **D** From the graph, when the current is 3.4 A, resistance is 2.4 Ω.
$P = I^2R = (3.4)^2 \times 2.4 = 27.7 = 28$ W (3 s.f.)

23 **D** Let the p.d. across the metre long wire be V.

By the potential divider principle, $\frac{V}{6.0} = \frac{7.5}{7.5+4.4} \Rightarrow V = 3.78$ V

$$\frac{x}{1.2} = \frac{1.0}{3.78} \Rightarrow x = 0.32\ \text{m}$$

24 **C** If there were no electronic component, the graph of I versus V is a straight line passing through the origin and having a gradient of $\frac{1}{4.0}\ \Omega^{-1}$.

Below 2.0 V, current only flows through the 4.0 Ω resistor, but when $V \geq 2.0$ V, the electronic component conducts a constant current. Hence, for $V \geq 2.0$ V, the straight line will be displaced vertically upwards by a constant value.

25 **A** By the right-hand grip rule, the magnetic field produced at the center of the coil by the current I_1 is directed perpendicularly into the paper. Then, using Fleming's Left-Hand rule, the force acting on the horizontal wire is perpendicular to it, in the plane of the page and pointing upwards towards the top edge of the page.

26 **B** As the magnet falls through the coil, the rate of change of magnetic flux linkage in the coil induces an emf and a current in the coil, producing a magnetic field that opposes the motion of the magnet to slow it down.

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The strength of this opposing magnetic field depends on the magnitude of the current in the coil I and the number of turns per unit length *n* of the coil, as given by $B = \mu_0 nI$.

Since length *L* of coil is fixed, *n* is higher for more turns *N* in the coil. The current in the circuit will be much higher if the electrical component is an ammeter since its resistance is much smaller than that in a voltmeter.

27 **B** Statement 1 is correct because higher energy electrons produce higher energy X-rays. Hence, the peak of the graph should be at a shorter wavelength (since $E = \frac{hc}{\lambda}$).

Statement 2 is incorrect because the intensity of the X-rays produced will be higher at each wavelength.
Statement 3 is also incorrect because the minimum wavelength of X-rays generated will be shorter since the electrons are more energetic (see reasoning for statement 1).

28 **D** Statement 1 is incorrect / Statement 2 is correct because the current is directly proportional to the rate of incidence of the photons, and not on the frequency of the photons.

Statement 3 is incorrect because the maximum KE_{max} of the emitted electrons depends only on the frequency *f* of the photons.

(Photoelectric equation: $hf = KE_{max} + \Phi$, where *h* is the Planck constant, *f* is frequency of photons, KE_{max} is max. *KE* of emitted electrons and Φ is the work function of the metal.)

29 **D** Balancing the mass numbers,

$$235 + 1 = X + 90 + Y\,(1)$$
$$236 - 90 = 146 = X + Y$$

Hence, the only pair that fits the condition above is D, where X = 144 and Y = 2.

(In fission reactions, the number of slow neutrons produced is 2 to 3.)

30 **C** Alpha radiation is harmful because of its high ionising power, and they are also easily absorbed by cells in the body (hence low penetrating power).

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Paper 2

1 **(a)** Upthrust U = weight of sea water displaced by the sphere

$$\begin{aligned} U &= \rho_{\text{sea water}} V_{\text{sphere}} g \\ &= 1020 \times \left(\frac{4}{3} \pi \times 0.180^3 \right) \times 9.81 \\ &= 244.41 \\ &= 244 \text{ N (3 s.f.)} \end{aligned}$$

(b) Since $U = \rho_{\text{sea water}} V_{\text{sphere}} g = \rho_{\text{sea water}} \left(\frac{4}{3} \pi \times r^3 \right) g$,

$$\frac{\Delta U}{U} = \left(\frac{\Delta \rho_{\text{sea water}}}{\rho_{\text{sea water}}} + 3 \frac{\Delta r}{r} \right)$$

$$\Delta U = \left(\frac{5}{1020} + 3 \times \frac{0.004}{0.180} \right) \times 244.41 = 17.49$$

$$= 20 \text{ N (1 sf)}$$

$$\therefore U \pm \Delta U = 240 \pm 20 \text{ N}$$

(c) Since the sphere is in equilibrium, vertical resultant force on the sphere is zero.

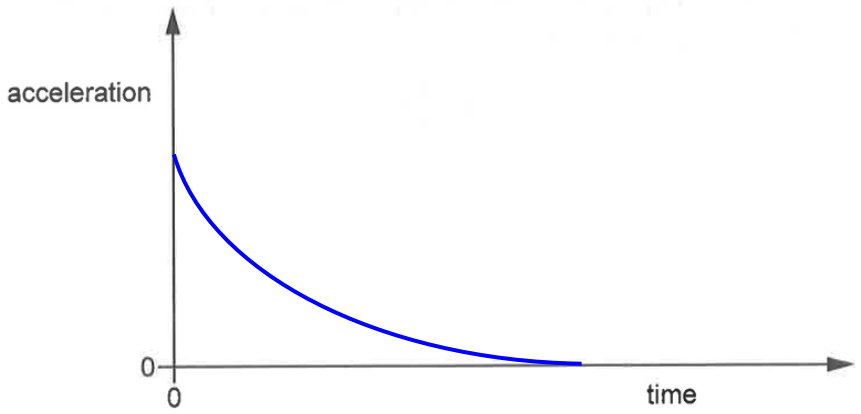
$$U = W + T$$

$$T = U - W = 244.41 - \left(85 \times \frac{4}{3} \pi \times 0.180^3 \times 9.81 \right)$$

$$= 244.41 - 20.37$$

$$= 224 \text{ N}$$

(d)



Explanation:
When the cable breaks, the sphere accelerates vertically upwards since $U > W$.

As the sphere accelerates upward, the drag force acting downwards on the sphere increases as its speed increases. By Newton's 2nd Law, $F_{net} = U - W - F_{drag} = ma$, the net force and hence acceleration decreases.

Since acceleration decreases, the velocity of the sphere increases at a decreasing rate. Thus drag force also increases at a decreasing rate which consequently results in the acceleration of the sphere decreasing at a decreasing rate.

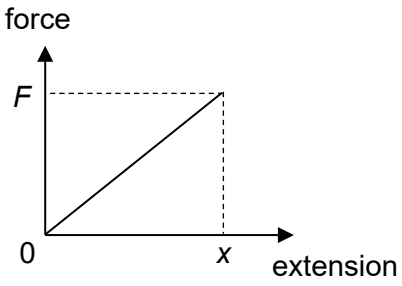
2 **(a)** **(i)** For a body in rotational equilibrium, the sum of all the clockwise moments about any axis must equal the sum of all the anticlockwise moments about the same axis.

(ii) The centre of gravity of a body is the point at which the weight of the body (or the resultant of the distributed gravitational attraction on the body) appears to act.

(b) **(i)** Since the spring obeys Hooke's Law, the load F on the spring is directly proportional to its extension x, hence,

$$F = kx$$

where k is the force constant of the spring.



The energy stored in the stretched spring W = area under the force-extension graph (which is the area of a right-angled triangle).

$$W = \frac{1}{2}Fx = \frac{1}{2}kx \cdot x$$
$$= \frac{1}{2}kx^2$$

(ii)
$$W = \frac{1}{2}kx^2$$
$$k = \frac{2W}{x^2} = \frac{2 \times 0.16}{0.081^2}$$
$$= 48.8 \text{ Nm}^{-1}$$

(c) **(i)** Let the length of the rod be L.

Taking moments about the hinge,

$$mg\left(\frac{L}{2}\right) = T\sin 38°(L)$$
$$m = \frac{2T\sin 38°}{g} = \frac{2 \times 48.8 \times 0.12 \times \sin 38°}{9.81}$$
$$= 0.735 \text{ kg}$$

(ii) When a load is added to the rod at X, the clockwise moment acting on the rod about the hinge increases. To maintain rotational equilibrium, the vertical component of the tension $T\sin 38°$ must increase, hence T increases in magnitude. Since the tension increases, the leftward horizontal component of the tension $T\cos 38°$ would

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also increase. To maintain translational equilibrium, the rightward horizontal component of force *H*, H_x increases.

By taking moments about X, since the anticlockwise moment due to the rod's weight remains unchanged, the vertical component of *H*, H_y remains the same. Hence, the magnitude of *H* must increase since $H = \sqrt{H_x^2 + H_y^2}$ due to the increase in H_x.

Since $\tan\theta = \frac{H_y}{H_x}$ and H_x increases, $\tan\theta$ and consequently θ decreases where θ is defined with respect to the horizontal rod.

3 **(a)** **(i)** Period $T = 0.40$ s

Angular frequency $\omega = \frac{2\pi}{T} = \frac{2\pi}{0.40} = 15.708 \approx 16$ rad s^{-1}

(ii) Amplitude $y_0 = \frac{18.0 - 6.0}{2} = 6.0$ mm

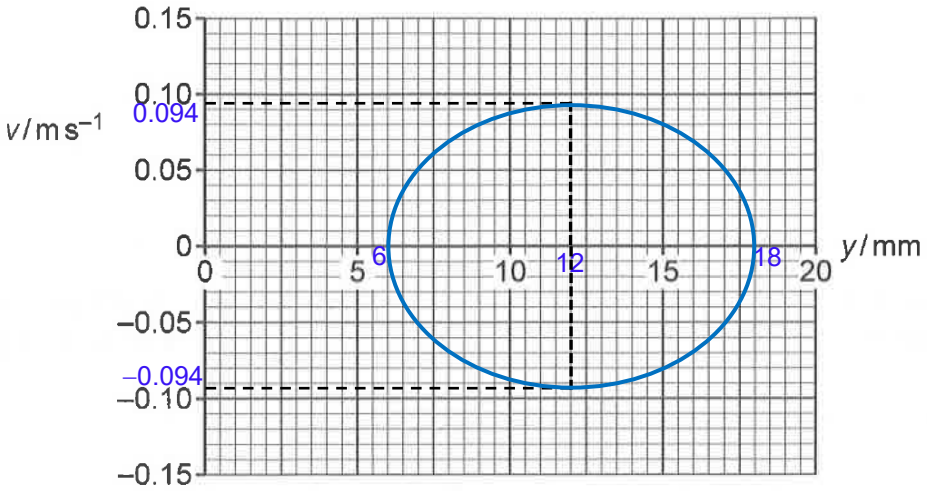
Magnitude of max. acceleration $a_0 = \omega^2 y_0 = (15.708)^2 \times 6.0 \times 10^{-3}$

$= 1.5$ m s^{-2}

(iii) Magnitude of max. velocity $v_0 = \omega y_0 = 15.708 \times 6.0 \times 10^{-3}$

$= 0.094$ m s^{-1}

(iv)



(v) At the top of the oscillation,

$mg - N = m\omega^2 x$

When the object loses contact with the platform at $x = A_0$, $N = 0$.

$mg = m\omega^2 A_0$

$A_0 = \frac{g}{\omega^2} = \frac{9.81}{16^2}$

$= 0.038$ m

(b) **(i)** Critical damping is the condition in which a damping force is applied so that the system returns to equilibrium in the shortest possible time without oscillating.

(ii) Critical damping removes energy from the system at the fastest rate without allowing oscillations to occur. Thus, critical damping is important because it ensures rapid dissipation of the disturbance energy without oscillation, giving optimal ride stability and comfort. It also leaves the car ready to respond to further bumps in the road quickly.

4 **(a)** The principle of superposition states that when two or more waves of the same kind meet at a point in space, the resultant displacement at that point is equal to the vector sum of the displacements of the individual waves at that point.

(b) Sound waves have wavelengths comparable to the width of a doorway, so they diffract noticeably when passing through it.

Light waves have wavelengths much smaller than the doorway width, so any diffraction is negligible and therefore not observed.

(c) **(i)** Using $d\sin\theta = n\lambda$,

$$d\sin\theta_x = 2\lambda_x \quad \text{..... (1)}$$
$$d\sin\theta_y = \lambda_y \quad \text{..... (2)}$$

Since both occur at point P, $\theta_x = \theta_y \Rightarrow \sin\theta_x = \sin\theta_y$,

$\therefore \lambda_y = 2\lambda_x$.

(ii) Since $\lambda_x = 340 \text{ nm} \Rightarrow \lambda_y = 680 \text{ nm}$.

The maximum number of maxima observable on one side of the principal (zeroth order) maxima is given by $n < \dfrac{d}{\lambda}$.

$$n_x < \frac{1}{500000 \times 340 \times 10^{-9}} = 5.88$$
$$\approx 5$$

$$n_y < \frac{1}{500000 \times 680 \times 10^{-9}} = 2.94$$
$$\approx 2$$

Hence the difference ΔN between the total number of maxima visible on the screen with laser X and total number visible with laser Y is

$$\Delta N = (2n_x + 1) - (2n_y + 1) = 2 \times 5 - 2 \times 2$$
$$= 6$$

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5 **(a)** It is the amount of electrical energy per unit charge that is converted from other forms of energy when charge passes through the source.

(b) **(i)** For a constant current, $I = \frac{Q}{t} = \frac{3.5 \times 10^{20} \times 1.6 \times 10^{-19}}{2.6 \times 60} = 0.3888$

$= 0.39$ A (2 s.f.)

(ii) $E = IR + Ir$

$R = \frac{E}{I} - r = \frac{1.5}{0.3888} - 0.45 = 3.408$

$= 3.4\ \Omega$ (2 s.f.)

(iii) $V_{\text{terminal}} = IR = 0.39 \times 3.4 = 1.3$ V

(iv) The effective resistance of R and the new resistor is lower than R as they are arranged in parallel. Hence the total circuit resistance decreases.

Hence the current in the circuit increases. This implies the potential difference across the internal resistor of the cell increases.

From $E = IR_{\text{effective}} + Ir$, the terminal p.d. across the cell (i.e. $E - Ir$) decreases.

6 **(a)** **(i)** From the power waveform, the period T is 50.0 ms.

$\therefore$ frequency $f = \frac{1}{T} = \frac{1}{50.0 \times 10^{-3}} = 20.0$ Hz

(ii) Mean power of supply $\langle P \rangle = \frac{P_0}{2} = \frac{0.90}{2} = 0.45$ W

$\langle P \rangle = I_{\text{rms}} V_{\text{rms}}$

$V_{\text{rms}} = \frac{\langle P \rangle}{I_{\text{rms}}} = \frac{\langle P \rangle}{I_0 / \sqrt{2}}$

$= \frac{0.45}{0.12 / \sqrt{2}} = 5.3033$

$= 5.3$ V

(b) For a sinusoidal AC source, the instantaneous power in a resistor is proportional to $\cos^2 \omega t$ hence the power–time graph is always positive and oscillates at twice the voltage frequency.

Since the average value of $\cos^2 \omega t$ over one full cycle is $\frac{1}{2}$, the mean power delivered is half of the maximum power.

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7 **(a)** They are nuclides that have one proton but different number of neutrons.

(b) **(i)**

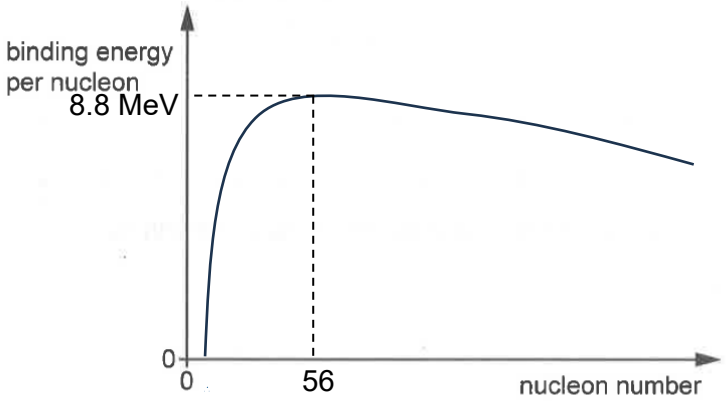


Fig. 7.1

(ii) This is because the maximum point of the graph in Fig. 7.1 is around iron-56.

When nuclei with nucleon numbers less than 56 are fused together, the nucleon number of the product increases and from the graph, binding energy per nucleon increases. Hence, energy is released in the fusion reactor and can be used as fuel.

(c) **(i)** Energy released $= 7.074 \times 4 - 2.827 \times 3 - 1.112 \times 2 = 17.591\,\text{MeV}$

(ii) Since the total momentum is zero, the momentum of the helium nucleus and the neutron are equal in magnitude and opposite in direction.

$E_k = \frac{1}{2}mv^2 = \frac{1}{2}\frac{(mv)^2}{m} = \frac{p^2}{2m}$ where p is the magnitude of momentum.

Since the two particles have the same p, their kinetic energy is inversely proportional to their respective masses. The mass of neutron is lower and hence it has higher kinetic energy.

(d) ${}^{1}_{0}\text{n} + {}^{6}_{3}\text{Li} \rightarrow {}^{3}_{1}\text{H} + {}^{4}_{2}\text{He}$

(e) Both nuclei have a charge of $+e = 1.6 \times 10^{-19}$ C. When they move across a magnetic field, each will experience a magnetic force of $Be v \sin\theta$ that is perpendicular to their velocities.

This changes the direction of motion of the charged nuclei.

(f) **(i)** $\frac{3}{2}kT = 105 \times 1.6 \times 10^{-16}$

$T = \frac{2 \times 105 \times 1.6 \times 10^{-16}}{3 \times 1.38 \times 10^{-23}} = 8.12 \times 10^{8}\ \text{K}$

(ii) The value is 5 orders of magnitude larger than the difference between the two scales which is 273.15. Adding or subtracting 273.15 to the answer in **(f)(i)** will not make any significant difference to the answer.

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(g) $\omega = 2\pi f = 2\pi \times \dfrac{225}{60} = 23.56 \text{ rad s}^{-1}$

$$3.75 \times 10^{9} = \frac{775000 \times r^{2} \times (23.56)^{2}}{4}$$

$$r = 5.90 \text{ m}$$

(h) **(i)** It refers to the resistance of an object to a change in its state of motion.

(ii) $\text{Time taken} = \dfrac{1500}{3.00 \times 10^{8}} = 5.00 \times 10^{-6} \text{ s}$

(iii) Number of tritium/deuterium nuclei = number of neutrons produced

$$= 4.8 \times 10^{17}$$

Since only 2.0% of the available fuel in the sphere is used,

$$\text{Number of each nuclide in the sphere} = \frac{4.8 \times 10^{17}}{0.02}$$

$$= 2.4 \times 10^{19}$$

$$\text{total mass of fuel} = 2.4 \times 10^{19} \times (2+3) \times 1.66 \times 10^{-27}$$

$$= 1.99 \times 10^{-7} \text{ kg}$$

(iv) The high-energy neutrons emitted during fusion are absorbed by the nuclei of the material of the container resulting in the formation of radioactive nuclides.

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Paper 3

1 **(a)**
$$\theta = \frac{2\pi}{3.4} \times 0.53$$
$$= 0.97944$$
$$= 0.979 \text{ rad}$$

(b) **(i)**
$$\omega = \frac{2\pi}{3.4}$$
$$= 1.8480$$
$$= 1.85 \text{ rad s}^{-1} \text{ (shown)}$$

(ii)
$$T\cos 10° = mg \quad - (1)$$
$$T\sin 10° = mr\omega^2 \quad - (2)$$
(2) / (1),
$$\tan 10° = \frac{r\omega^2}{g}$$
$$r = \frac{9.81\tan 10°}{1.85^2}$$
$$= 0.50541$$
$$= 0.505 \text{ m}$$

(c) Work done is the product of force and displacement in the direction of force.

The weight acts vertically while the bob's displacement is horizontal, so weight has no component in the direction of motion and does no work.

The tension acts along the string toward the pivot, which is perpendicular to the bob's velocity, so tension also has no component in the direction of motion and does no work.

2 **(a)** Newton's third law of motion states that if a body A exerts a force on body B, then body B exerts an equal and opposite force on body A.

(b)

	direction	magnitude	body on which the body acts
Newton's third law pair with weight	upwards	W	Earth
Newton's third law pair with normal contact force	downwards	N	floor of elevator

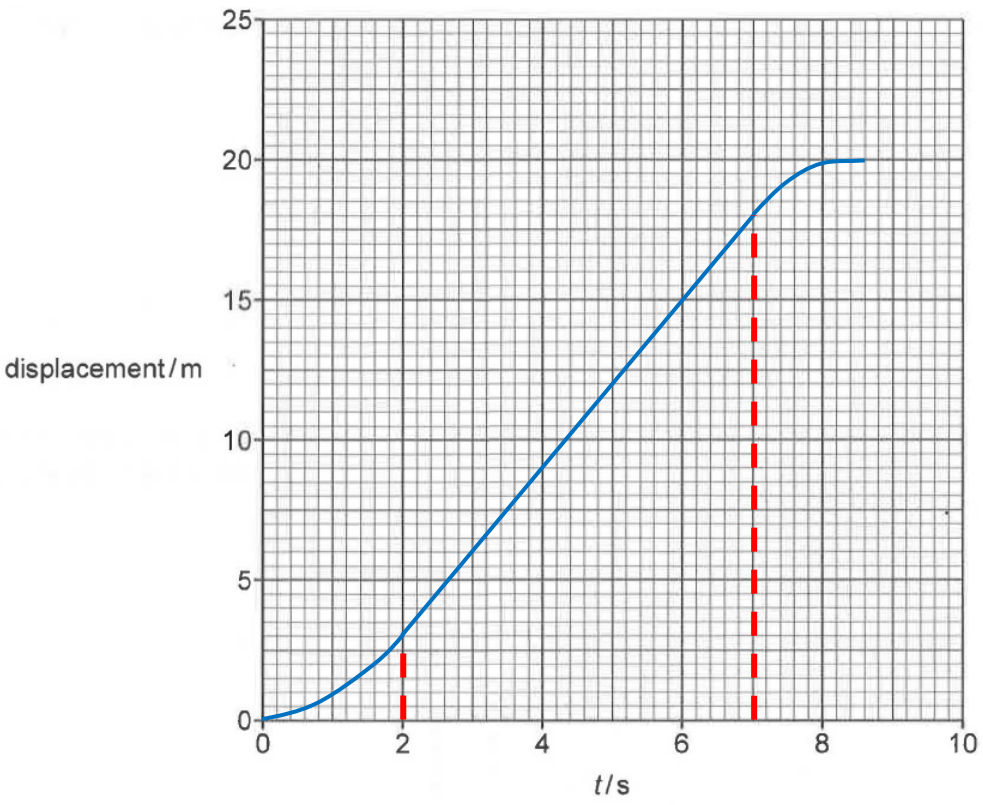
(c) **(i)**

$$ma = W - N$$
$$(65)(1.5) = (65)(9.81) - N$$
$$N = 540.15$$
$$= 540 \text{ N}$$

(ii) **(A)**

$$s = ut + \frac{1}{2}at^2$$
$$= 0 + \frac{1}{2}(1.5)(2.0)^2 = 3.0 \text{ m (shown)}$$

(B) & (iii)



from 2.0 to 7.0 s,

$v = u + at = 0 + (1.5)(2.0) = 3.0 \text{ m s}^{-1}$

$s = vt = (3.0)(5.0) = 15 \text{ m}$

The gradient of the displacement-time graph at t = 0 and final time must be zero as the speed is zero.

As question did not state the value of deceleration of the lift after 7.0 s, the time at which the lift reaches 20 m can be any time after 7.0 s.

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3 **(a)** **(i)** The internal energy of a system is the sum of the microscopic kinetic energy, due to the random motion of the molecules, and the microscopic potential energy, associated with the intermolecular forces of the system.

(ii) The kinetic energy of the water molecules remains constant since temperature remains constant.

When melting, energy is required to weaken the bonds between the water molecules and move them further apart. This increases its potential energy (potential energy associated with these intermolecular forces becomes less negative).

Hence, the internal energy increases.

(b) Since there is still ice in the beaker when thermal equilibrium is reached, the final temperature of the system must be 0° C.

$$\begin{aligned}
\text{heat gained} &= \text{heat lost} \\
m_{\text{melted ice}}L + m_{\text{ice}}c_{\text{ice}}\left(0-(-15)\right) &= m_{\text{water}}c_{\text{water}}\left(100-0\right) + m_{\text{glass}}c_{\text{glass}}\left(28-0\right) \\
m_{\text{melted ice}}\left(334\right) + \left(120\right)\left(2.05\right)\left(15\right) &= \left(90\right)\left(4.18\right)\left(100\right) + \left(170\right)\left(0.84\right)\left(28\right) \\
m_{\text{melted ice}} &= 113.56 \\
&= 114 \text{ g}
\end{aligned}$$

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4 **(a)** Light is a transverse wave which can be polarised, but sound is a longitudinal wave which cannot be polarised.

(b) **(i)** 90°

(ii) Using Malus law,

$I_Q = I_0 \cos^2 \theta$

$I_P = I_Q \cos^2 (90° - \theta) = I_0 \cos^2 \theta \cos^2 (90° - \theta)$

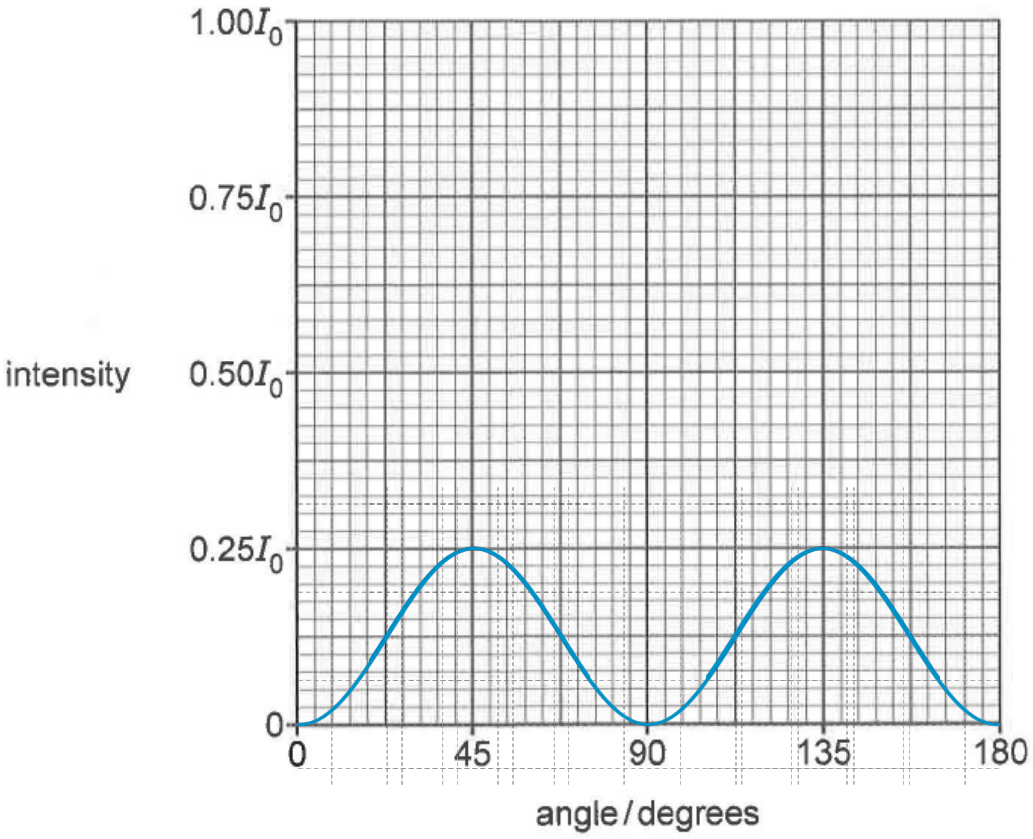
$= I_0 \cos^2 \theta \sin^2 \theta$

$= 0.25\, I_0 \sin^2 2\theta \qquad (\sin 2\theta = 2 \sin \theta \cos \theta)$

At 0°, 90°, 180°, $I_p = 0$

At 45°, $I_p = 0.25\, I_0$

At 90°, 135°, $I_p = 0.25\, I_0$



(c) **(i)** $I \propto \frac{1}{r^2}$

$\frac{I}{I_0} = \left(\frac{50}{10}\right)^2$

$I = 25 I_0$

(ii) $I \propto A^2$

$\frac{25 I_0}{I_0} = \left(\frac{A}{A_Q}\right)^2$

$A_Q = 0.20 A$

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5 **(a)** An emission line spectrum consists of bright lines on a dark background.

Each line corresponds to light of a specific wavelength emitted when electrons in atoms de-excite between discrete energy levels, releasing photons of specific energies corresponding to the difference in the energy levels (and hence specific wavelengths).

(b) **(i)**

$$\text{KE of electron} = \frac{1}{2}m_e v^2$$
$$= \frac{1}{2}\left(9.11\times10^{-31}\right)\left(1.93\times10^{6}\right)^2$$
$$= 1.6967\times10^{-18}\ \text{J}$$
$$= 10.604\ \text{eV}$$

Assuming all the energy of the incoming electron is transferred in a head-on collision, the hydrogen electron can be excited only up to the −3.40 eV level, because it can absorb energy only in discrete amounts corresponding to its allowed energy levels.

Therefore maximum energy that can be absorbed by the electron
= –3.40 – (–13.59) = 10.19 eV

(ii)

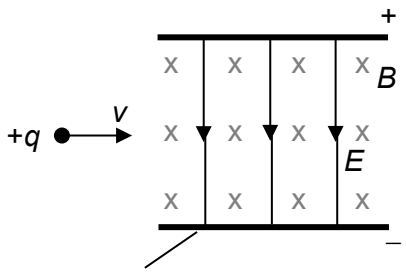
$$\frac{hc}{\lambda} = E_{\text{max}}$$
$$\lambda = \frac{\left(6.63\times10^{-34}\right)\left(3.00\times10^{8}\right)}{10.19\times1.6\times10^{-19}}$$
$$= 1.2199\times10^{-7}$$
$$= 1.22\times10^{-7}\ \text{m}$$

(iii)

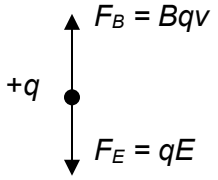
$$\text{energy of deflected electron, } E = 1.6967\times10^{-18} - 10.19\times1.6\times10^{-19}$$
$$= 6.63\times10^{-20}\ \text{J}$$
$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2m_e E}}$$
$$= \frac{6.63\times10^{-34}}{\sqrt{2\times9.11\times10^{-31}\times6.63\times10^{-20}}}$$
$$= 1.9076\times10^{-9}$$
$$= 1.91\times10^{-9}\ \text{m}$$

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6 (a)



parallel electric plates



forces acting on the charge particle

A charged particle enters a region where the electric field E and magnetic field B are applied perpendicular to each other in an orientation such that the electric force and the magnetic force on the particle act in opposite directions.

The electric force on the particle is qE, and the magnetic force is Bqv where q is the charge of the particle and v is the velocity of the particle.

Only particles with $v = \frac{E}{B}$ experience equal and opposite forces ($qE = Bqv$), and therefore pass through undeflected by travelling at a constant velocity (since $F_{net} = 0$).

Particles with speeds faster or slower than $\frac{E}{B}$ are deflected, and therefore do not pass through the selector.

This allows for the selection of particles with only a specific velocity to pass through.

(b) $Bqv = qE$

$$B = \frac{E}{v} = \frac{V/d}{v}$$
$$= \frac{410/0.030}{8.6 \times 10^5}$$
$$= 0.015891$$
$$= 0.0159 \text{ T}$$

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7 **(a)** $A = \lambda N$

$$= \left(\frac{\ln 2}{t_{1/2}}\right)\left(\frac{m}{M_R} \times N_A\right)$$

$$= \left(\frac{\ln 2}{14.3 \times 24 \times 60 \times 60}\right)\left(\frac{3.0 \times 10^{-6}}{32} \times 6.02 \times 10^{23}\right)$$

$$= 3.1662 \times 10^{10}$$

$$= 3.17 \times 10^{10} \text{ Bq}$$

(b) In the decay, the total energy (1.71 MeV) is shared between the electron and the antineutrino in a way such that total momentum is conserved.

Since the antineutrino can take different amounts of energy and momentum, the beta particle will have a continuous range of energy up to a maximum of 1.71 MeV.

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8 **(a)** Magnetic flux is defined as the product of the magnetic flux density (through an area) and the cross-sectional area perpendicular to the direction of the magnetic flux density.

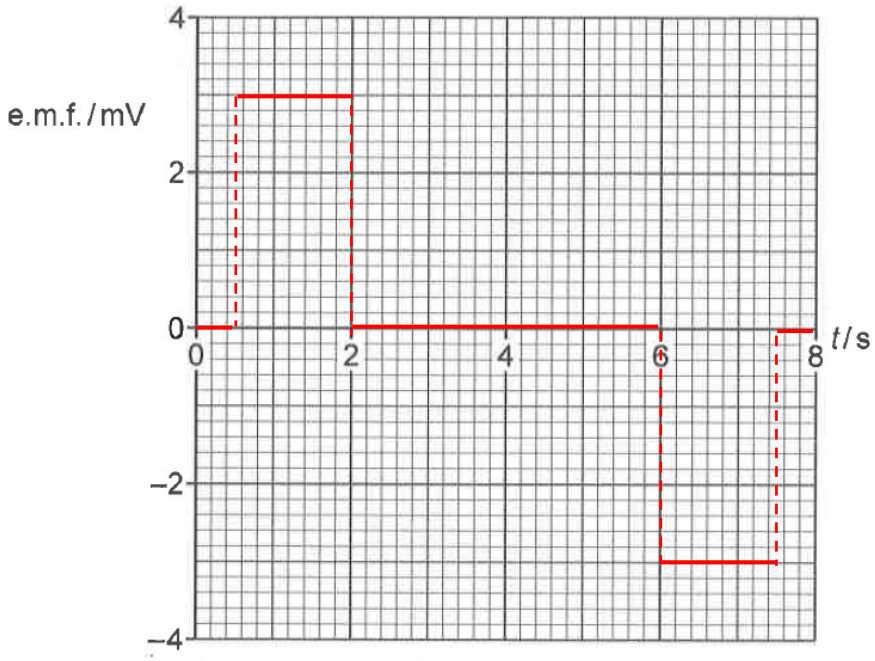
(b) **(i)** Induced e.m.f. $= NBlv$

$$= (300)\left(4.2\times10^{-3}\right)(0.060)(0.040)$$

$$= 3.0\times10^{-3}\ \text{V}$$

$$= 3.0\ \text{mV (shown)}$$

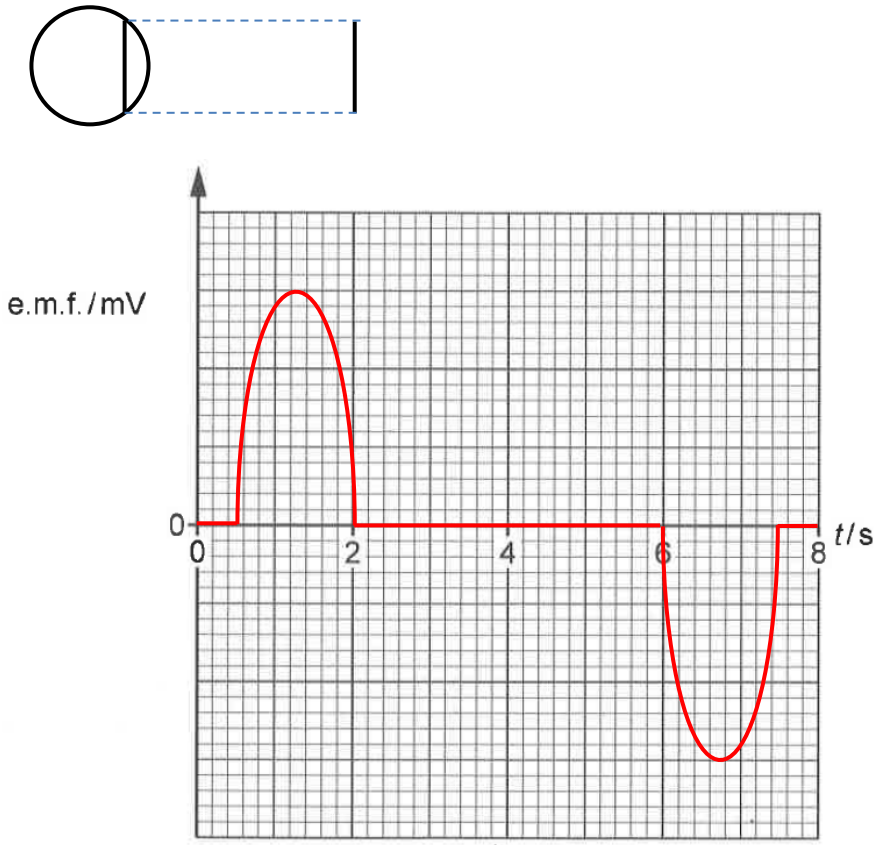
(ii)



The coil enters the *B*-field at time t = 0.5 s.
Time duration for coil to be completed in the *B*-field = 6 / 4 = 1.5 s.
Time duration for coil to move another 16 cm = 16 / 4 = 4.0 s.
Time duration for coil be completed out of *B*-field = 6 / 4 = 1.5 s.

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(c)

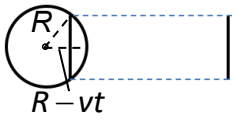


The coil enters the *B*-field at time t = 0.5 s.
Time duration for ½ the coil to be completed in the *B*-field = 0.75 s.
Time duration for the coil to be completed in the *B*-field = 6 / 4 = 1.5 s.
Time duration for coil to move another 16 cm = 16 / 4 = 4.0 s.
Time duration for coil be completed out of *B*-field = 6 / 4 = 1.5 s.

When coil moves into *B*-field, the induced e.m.f. is proportional to the instantaneous length of the chord of the arc that is in the *B*-field which varies non-linearly with *t*.

Additional note:
When coil is within the uniform B-field after time *t*,

$$\text{Chord length} = 2\sqrt{R^2 - (R - vt)^2}$$
$$= 2\sqrt{R^2 - R^2 + 2Rvt - v^2t^2}$$
$$= 2\sqrt{2Rvt - v^2t^2}$$



(d) When the ends of the coil are connected together, the closed circuit allows an induced current to flow when there is a rate of change of magnetic flux linkage through the coil.

By Lenz's Law, the induced current produces an effect to oppose the change in magnetic flux linkage. This effect produces an opposing force to the motion of the coil when it moves into the *B*-field.

Hence, work done by an external force is needed in order for the coil to continue to move with a constant speed, without which the coil will slow down due to the opposing force.

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(e) (i) Current through the solenoid $= \frac{V}{R} = \frac{1.2}{0.27} = 4.4$ A

Magnetic flux density at the centre of solenoid $= \mu_0 nI$

$4.2 \times 10^{-3} = \left(4\pi \times 10^{-7}\right) n (4.4)$

$n = 752$ (where $n =$ number of turns per unit length)

Length of wire needed to make one turn $=$ circumference of solenoid

$= \pi d$

$= 0.030\pi$

Number of turns possible using 12 m of wire $= \frac{12}{0.030\pi} = 127.3$

$n = \frac{N}{L}$

$752 = \frac{127.3}{L}$

$L = 0.169$ m

(ii) The solenoid has a diameter of only 3.0 cm while the coil is 6.0 cm. The coil will not be able to enter the solenoid.

The magnetic flux density of the solenoid is not uniform.

(iii) $R = \frac{\rho l}{A}$

$0.27 = \frac{\left(1.7 \times 10^{-8}\right)(12)}{\pi r^2}$

$r = 4.90 \times 10^{-4}$ m

$d = 2r = 9.80 \times 10^{-4}$ m

$= 0.98$ mm

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9 (a) (i) A *field of force* is a region of space in which an when object placed in that region, with certain properties, will experience a force associated with the field.

(ii) Similarities:
Both gravitational fields and electric fields that are produced from a point source (point mass or point charge) follow an inverse square law (i.e. the field strength is inversely proportional to the square of the distance from the point source).

The gravitational and electric fields always act in the direction of decreasing potential since $g = -\frac{d\phi}{dx}$ and $E = -\frac{dV}{dx}$.)

Differences:
Electric fields act on objects with electric charge, while gravitational fields act on objects with mass.

An object with a mass will always feel a gravitational force in the direction of the gravitational field, while an object with a charge may feel an electric force in either the same or opposite direction to the electric field (depending on the sign of the charge).

(b) (i)

$$g = \frac{GM}{r^2}$$

$$2.71 = \frac{(6.67 \times 10^{-11})(9.63 \times 10^{22})}{r^2}$$

$$r = 1.54 \times 10^{6} \text{ m}$$

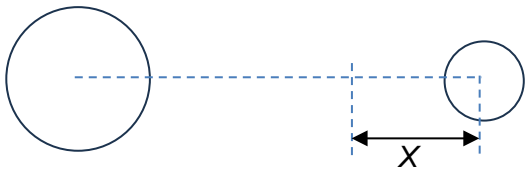
(ii) The resultant gravitational field on the surface of the moon is due to the vector sum of the gravitational field due to the moon and the planet (and not due to the moon alone).

From **(b)(i)**, the diameter of the moon is not negligible compared to the distance between the planet and the moon.

Hence, the gravitational field due to the planet on a point on the moon facing the planet would be significantly different from the gravitational field due to the planet on a point on the moon further away from the planet.

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(iii) **(A)**



$$g_P = g_M$$

$$\frac{GM_P}{r^2} = \frac{GM_M}{r^2}$$

$$\frac{(9.04\times10^{24})}{(2.67\times10^{8}-x)^2} = \frac{(9.63\times10^{22})}{(x)^2}$$

$$x = 2.50\times10^{7}\,\text{m}$$

Therefore distance from surface of the moon = $2.50\times10^7 - 1.54\times10^6$
= 2.35×10^7 m

(B) Gravitational potential is a scalar quantity. The gravitational potential at the null point is the scalar addition of the contributions of gravitational potentials from both the planet and the moon.

And since the contributions are both negative, the gravitational potential at the null point is therefore not zero.

(C)

$$\begin{aligned}\phi_{total} &= \phi_P + \phi_M \\ &= -\frac{GM_P}{r_1} - \frac{GM_M}{r_2} \\ &= -\frac{(6.67\times10^{-11})(9.04\times10^{24})}{2.67\times10^{8}-2.50\times10^{7}} - \frac{(6.67\times10^{-11})(9.63\times10^{22})}{2.50\times10^{7}} \\ &= -2.75\times10^{6}\,\text{J kg}^{-1}\end{aligned}$$

(c) Since there is a point between the charges where the electric potential is zero, it implies that the charges are of opposite signs.

Hence, on a line between these two charges, the electric fields due to each of the charges point in the same direction, and therefore it is not possible for the resultant field strength to be zero.

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